DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY OF BARISHAL
FINAL EXAMINATION

Course Title: Discrete Mathematics
Course Code: CSE-1203
1st Year, 2nd Semester
Session: 2018-19

Marks: 60

Time: 3 hours

Answer any five Questions from the followings. Parts of the same question should be answered consecutively.

1. a) When three professors are seated in a restaurant, the hostess asks them: "Does everyone want coffee?" The first professor says: "I do not know." The second professor then says: "I do not know." Finally, the third professor says: "No, not everyone wants coffee." The hostess comes back and gives coffee to the professors who want it. How did she figure out who wanted coffee? [3]

   b) Show that $(a \to c) \wedge (b \to c)$ and $(a \vee b) \to c$ are logically equivalent using truth table. [5]

   c) Let $R(p, q, r)$ denote the statement $p - q = r$. What are the truth values of the propositions $R(-1, 2, -3)$ and $R(-1, -2, -3)$? Let Q(x) denotes the statement x=x+1. What is the truth value of the qualification $\exists x Q(x)$, where domain consists of all real numbers? [4]

2. a) Define set with example. What is the power set of {a, b, c, 2} and Cartesian products of {1, 2, 3} and {x, y, z}. [4]

   b) Verify the proposition $p \vee \neg (p \wedge q)$ is tautology. [4]

   c) Let A={1, 2, 3, 4}. Determine the truth value of each of the following statements. [4]
      (i) $(\exists x \in A)(x+3=10)$ (ii) $(\forall x \in A)(x+3<10)$
      (iii) $(\exists x \in A)(x+3<5)$ (iv) $(\forall x \in A)(x+3<7)$

3. a) Find the domain and range of the following function. Note that in each case, to find the domain, determine the set of elements assigned values by the function. [4]
      i. *The function that assigns to each bit string twice the number of zeros in that string.*
      ii. *The function that assigns to each bit string the number of ones in the string minus the number of zeros in the string.*

   b) Find $f \circ g$ and $g \circ f$, where $f(y) = y^2 + 1$ and $g(y) = y + 2$, are function from R to R. [4]

   c) Draw the graph of the function $f(x) = 1 - x^2$ from **Z** to **Z**. [2]

   d) Find the inverse function of $f(x) = x^3 + 1$. [2]

4. a) Consider the following relation R on the set A = {1, 2, 3}: [4]
      R = {(1, 2), (2, 3), (3, 3)}
      i. Find reflexive(R)
      ii. Find symmetric(R) and
      iii. Find transitive(R)

   b) Given A={1, 2, 3, 4} and B={x, y, z}. Let R be the following relation from A to B: [4]
      R={(1,y), (1,z), (3,y), (4,x), (4,z)}.
      (i) Determine the matrix of the relation.
      (ii) Find the inverse relation of R.
      (iii) Determine the domain and range of R.

   c) Let A={1, 2, 3, 4}. Consider the following relation in A: R= {(1,1), (2,2), (2,3), (3,2), (4,2), (4,4)} [4]
      (i) Draw its directed graph.
      (ii) Is R transitive relation?
      (iii) Find $R^2 = RoR$.

5. a) Find formulae for the sequence with the following first five terms: $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$. [3]

   b) Compute the following double sum: $\sum_{i=0}^{2} \sum_{j=0}^{3} i^3 j^2$. [2]

   c) Let $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 4 \\ 1 & 3 \end{bmatrix}$, does $AB = BA$? [4]

   d) Let $A = \begin{bmatrix} -1 & 2 \\ 1 & 3 \end{bmatrix}$, find A^{-1}. [3]

6. a) Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there? [3]

   b) Briefly explain the Pigeonhole principle with example. Show that if there are 30 students in a class, then at least two have last names that begin with the same letter. [3]

   c) Determine all the rows of a Pascal's Triangle, where $n = 0$ to $n = 10$. [3]

   d) What is the probability that a randomly selected day of a leap year is in April? [3]

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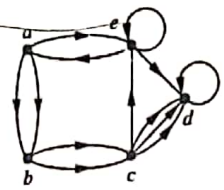
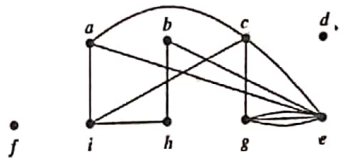
7. a) What do you mean by a *graph*? Draw two graph models. [3]

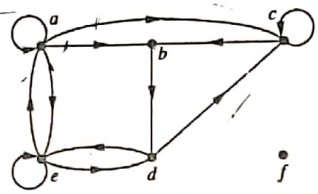
b) Distinguish between a Simple graph and a Multigraph. Determine the type of the following graph. How can it be converted to the opposite graph type? [3]



c) What do you mean by the neighborhood of a vertex? What are the neighborhoods of each vertex in the following graph? [3]

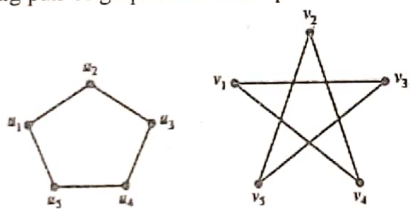


d) Find the in-degree and out-degree of each vertex in the following graph with directed edges. [3]

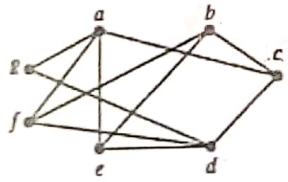


8. a) What do you mean by a complete graph? Draw a K_8 graph. What is the degree of a vertex of the K_8 graph you have drawn? [4]

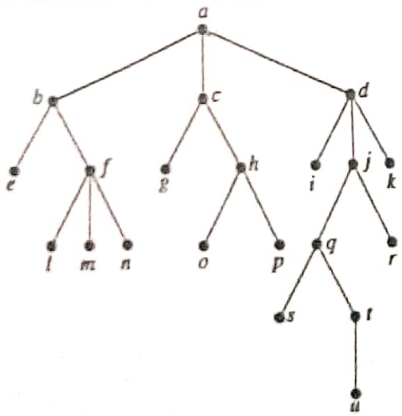
b) Determine whether the following pair of graphs are isomorphic. [2]



c) Define a bipartite graph. Determine whether the following graph is a bipartite graph. If it is a bipartite graph, color the graph using two different colors following the conditions of bipartite graph. [3]



d) Answer the following questions about the rooted tree illustrated below. [3]



i. Which vertices are leaves?
ii. Which vertices are children of f?
iii. Which vertex is the parent of h?
iv. Which vertices are siblings of o?
v. Which vertices are ancestors of m?
vi. Which vertices are descendants of b?

Good Luck!!!

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